

Automation of communication in Health and Social Care

(with a focus on services and support provided to older adults)

Health care refers to patients' medical and physical well-being, with staff providing services such as preventive care, diagnosis and treatment, and rehabilitation. However, social care refers to the social and emotional well-being of patients, providing services such as counselling, support for living, and daily assistance. Health and social care workers often work in professional settings such as hospitals, clinics, care homes, and community health centres.



monitor

To track and access older adults' symptoms and performance

- answers FAQs about medications, diet, conditions...
- delivers educational modules on diet, conditions,...
- provides tips on nutrition, physical activity, sleep, hygiene,...
- provides accessible explanations of medical terms, results,...
- helping older adults and families understand treatment pathways

- offering routine check-ins
- collects structured self-reports on pain, fatigue, appetite
- uses interactive exercises to monitor cognitive performance
- analyze sentiment in messages, tone of voice or keywords

health information provider

To provide health information for older adults' learning and decision making

- real-time multilingual translation during care conversations
- providing speech recognition for non-standard speech

cultural and language translator

To provide multilingual, inclusive, and culturally appropriate communication.

- offering culturally appropriate communication
- simplifying medical jargon into plain language
- providing multilingual documents and explanations

health coach

To train older adults and help them adopt healthy habits and living skills

- positive reinforcement
- encouragement during setbacks
- gives small 'nudges' to encourage healthy behaviour
- records routines, progress, achievements; identifies deviations
- step-by-step guidance for exercises, eg. memory games
- goal setting and progress visualization (gamification, badges)
- adjust recommendations to age, health, personal preferences

organise and coordinate

To plan, document, coordinate and evaluate care and health measures.

- checking in patients
- pre-screen symptoms
- appointment coordination (scheduling, rescheduling and reminders)
- step-by-step guidance through paperwork
- provides summaries after doctors' appointments
- routes patients to appropriate provider (triage support)
- facilitates communication between different providers or caregivers

disease health management assistant

To help older adults manage specific diseases

- collecting symptoms and health metrics (can integrate with medical devices)
- medication reminders
- detects dangerous patterns
- suggestions tailored to the condition

therapist

To deliver therapy to older adults for mental illness treatment

- immediate support in critical moments (24/7 availability)
- personalised interactive CBT tools
- 'psychological safe space' to open up
- assessment of mental health symptoms (chat content → keywords, sentiment,...)
- 'bridge' between patient and therapist (option to share chatbot conversation on sensitive topics)
- therapist integration (summaries, highlights important issues)



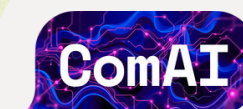
companion

To be with older adults in their everyday life and enhance their emotional status

- offering small talks and greetings
- offering check-ins (How are you doing?)
- suggesting entertaining and engaging activities
- simplifies video calls, messages and photo sharing with loved ones
- notifies caregiver/ family about moods, changes
- stimulate presence



indirect service and system-related



Researching the Automation of Societal Communication